The exact two-thirds scale for the Euclidean deviation of small-parameter LQG geodesics

Abstract

Let M_γ be the maximum Euclidean distance from $[0,1]$ attained by the γ-Liouville quantum gravity geodesic from 0 to 1 in a whole-plane Gaussian free field. We prove that $M_\gamma/\gamma^{2/3}$ is tight and is uniformly bounded away from zero in probability as $\gamma \to 0$. More precisely, its first moment is bounded, its asymptotic upper tail is at most $C(A^{-3} + A^{-4})$, and its lower tail is at most $Ca^{3/5}$. The proof works directly with the LQG metric. It does not require a limiting variational problem or uniqueness of a subsequential limit. The upper bound uses Gaussian occupation estimates and localized conformal perturbations; the lower bound uses disjoint Möbius images of a thin tube.

1 Statement and conventions

Let h be a whole-plane Gaussian free field (GFF), normalized initially so that its average on $\partial\mathbb{D}$ is zero. Our Dirichlet inner product is

$$(f,g)_\nabla = \frac{1}{2\pi}\int_{\mathbb{C}} \nabla f \cdot \nabla g \,\mathrm{d}z. \tag{1.1}$$

The covariance kernel is consequently

$$G(z,w) = \log\max\{1,|z|\} + \log\max\{1,|w|\} - \log|z-w|. \tag{1.2}$$

Write D_h^γ for the γ-LQG metric, and P_γ for its almost surely unique geodesic from 0 to 1, regarded as an unparametrized curve. Set

$$M_\gamma = \max_{z\in P_\gamma} \operatorname{dist}(z,[0,1]), \qquad \xi = \frac{\gamma}{d_\gamma}, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}, \quad p_\gamma = \xi Q.$$

Multiplying the metric by a positive constant, even a random one, changes neither P_γ nor M_γ.

Theorem 1.1. *There exist deterministic constants $C < \infty$ and $\gamma_0 > 0$ such that*

$$\mathbb{E}M_\gamma \le C\gamma^{2/3}, \qquad 0 < \gamma < \gamma_0. \tag{1.3}$$

Moreover, for every $A \ge 1$,

$$\limsup_{\gamma\to 0} \mathbb{P}\Big[M_\gamma > A\gamma^{2/3}\Big] \le C(A^{-3} + A^{-4}), \tag{1.4}$$

and, uniformly over $0 < \gamma < \gamma_0$ and $0 < a \le 1$,

$$\mathbb{P}\Big[M_\gamma \le a\gamma^{2/3}\Big] \le Ca^{3/5}. \tag{1.5}$$

In particular, for every $\varepsilon > 0$ there exist $a > 0$, $A < \infty$, and $\gamma_\varepsilon > 0$ such that

$$\mathbb{P}\Big[a \le \frac{M_\gamma}{\gamma^{2/3}} \le A\Big] \ge 1 - \varepsilon, \qquad 0 < \gamma < \gamma_\varepsilon. \tag{1.6}$$

For each $r \in (0, 3/5)$,

$$\sup_{0<\gamma<\gamma_0} \mathbb{E}\left[\left(\frac{M_\gamma}{\gamma^{2/3}}\right)^{-r}\right] < \infty. \tag{1.7}$$

Thus "does not collapse to zero" holds in the stronger sense of (1.6), not merely as failure of convergence to zero in probability. Every subsequential weak limit is finite and strictly positive almost surely. Also, for each $\varepsilon > 0$,

$$\mathbb{P}\Big[\gamma^{2/3+\varepsilon} \le M_\gamma \le \gamma^{2/3-\varepsilon}\Big] \longrightarrow 1. \tag{1.8}$$

These statements do not assert convergence to a unique law, or exclude a positive deterministic subsequential limit.

We use the established dimension estimate only through

$$\xi \sim \gamma/2, \qquad |p_\gamma - 1| \le C\xi. \tag{1.9}$$

For example, this follows from $2 < d_\gamma \le 2 + \sqrt{2}\gamma + \gamma^2/2$ [1, Theorem 1.2]. The sharper dimension expansion proved in our earlier work is consistent with (1.9) but is not needed for this proof.

Throughout, C, c denote positive deterministic constants which may change from line to line. A subscript records a parameter on which a constant may depend. A statement about each fixed moment order allows the small-γ threshold to depend on that order. When a common threshold for growing moment orders is required, it will be proved explicitly.

2 Metric inputs and uniform occupation estimates

2.1 Established LQG inputs

We use metric locality and Weyl scaling from [2, Section 1.2 and Remark 1.5], deterministic-pair geodesic uniqueness from [4, Lemma 2.2], and full conformal covariance from [3, Theorem 1.3]. In particular, for a conformal map $\phi : V \to U$, with inverse ψ,

$$D_h(z, w; V) = D_{h\circ\psi + Q\log|\psi'|}(\phi(z), \phi(w); U). \tag{2.1}$$

The semicolon denotes the internal metric, defined by restricting paths to the indicated domain. We also use bounded compactness of the metric, as recorded in [2, Section 1.5], when taking limits of geodesics with uniformly bounded lengths.

Let $\widehat{D}_h$ be normalized by the deterministic median of $D_h(0, 1)$. We use the following results of [5]: locally uniform Euclidean convergence, Theorem 1.7; fixed positive internal-diameter moments, Lemma 5.4; and exponential near-independence for separated annular events, Lemma 3.4. In the diameter lemma the available orders are $q < c_1\sqrt{M}$, where the auxiliary parameter M is chosen first. We explain the normalization transfer and all further moment estimates below. No assertion uniform in all q is imported from that lemma.

2.2 Fixed moments and localization

Let $P : [0,1] \to \mathbb{C}$ parametrize the 0-to-1 geodesic at constant $\widehat{D}_h$-speed, let $\mu = P_* \mathrm{d}t$, and let λ be uniform measure on $[0,1]$. For a finite-energy measure ν, write $\|\nu\|_G^2 = \iint G(z,w)\nu(\mathrm{d}z)\nu(\mathrm{d}w)$.

Lemma 2.1. *For each fixed $q < \infty$, the distance $\widehat{D}_h(0,1)$, its reciprocal, and the escape radius $\mathcal{R} = \max\{2, \sup_t |P(t)|\}$ have uniformly bounded qth moments for sufficiently small γ. There are a deterministic $K < \infty$ and a random C_* with a uniformly bounded positive moment such that*

$$\widehat{D}_h(u,v) \le C_*(1+R)^K |u-v|^{1/2}, \quad u,v \in B_R(0), \quad |u-v| \le 1. \tag{2.2}$$

The moment order of C_* can be chosen arbitrarily in advance. There are $b, C, \gamma_0 > 0$ for which

$$\sup_{0<\gamma<\gamma_0} \mathbb{E} \exp\{b \|\mu\|_G^2\} \le C. \tag{2.3}$$

Proof. First compare the annular quantile normalization in [5, Lemma 5.4] with median normalization. A fixed sufficiently fine polygon around the circle of radius $3/2$ can be joined by finitely many metric geodesics. Local Euclidean convergence and the ellipse inequality imply, with probability tending to one, that these geodesics stay in the annulus $\{1 < |z| < 2\}$, are homotopic there to the polygon edges, and have bounded total $\widehat{D}_h$-length. To justify localization, a first exit from a fixed neighborhood of an edge has Euclidean detour bounded below, whereas uniform metric error tends to zero. Their concatenation winds around the annulus. Hence the fixed high annular quantile is at most a constant times the point-distance median. The source diameter moments therefore give, for every prescribed fixed q,

$$\mathbb{E}\widehat{D}_h(0,1)^q \le C_q \tag{2.4}$$

and the corresponding compact internal-diameter moments.

Here is the inverse-moment argument. Fix small deterministic $c > 0$ and let E_r be the event

$$\widehat{D}_{h-h_r(0)}(\text{across } \{r/4 < |z| < r/2\}) \ge c r^{p_\gamma},$$

where the crossing length is internal to the annulus. This event is measurable from the field on that annulus with its r-circle average removed. By affine covariance its probability is independent of r; by local Euclidean convergence it tends to one as $\gamma \to 0$. The annular near-independence lemma, applied at radii 8^{-k}, implies that, for each prescribed A, the probability of no good annulus among the first n is at most e^{-An} for sufficiently small γ. On the event $\min_{k \le n} \xi h_{8^{-k}}(0) \ge -n$, any good annulus yields

$$\widehat{D}_h(0,1) \ge c \exp\{-(2\log 8 + 1)n\},$$

using $p_\gamma \le 2$. The circle-average process is Brownian in logarithmic radius, so the exceptional probability is at most $n \exp\{-n/(2\xi_0^2 \log 8)\}$ for $\xi \le \xi_0$. Choosing A large and then ξ_0 small proves every fixed inverse moment. For escape, use the outward annuli 8^k, $n/2 \le k \le n$, in reverse order, with the circle-average lower threshold $\xi h_{8^k}(0) \ge -(\log 8)n/8$. Since $p_\gamma \ge 1/2$, except with probability $e^{-An/2} + ne^{-cn/\xi_0^2}$, one crossed annulus costs at least $c8^{n/8}$. Combine this with (2.4) to obtain every fixed moment of $\mathcal{R}$.

Affine covariance and Cauchy–Schwarz give, for fixed q, small r, and centers in a compact set,

$$\mathbb{E}\,\mathrm{diam}_{\widehat{D}_h}(B_r(z); B_{2r}(z))^q \le C_q r^{p_\gamma q - q^2\xi^2}. \tag{2.5}$$

Fixed enlargements of these balls are harmless. The centered field has the canonical marginal law and the circle-average variance is $\log(1/r) + O(1)$; independence of these two quantities is not used.

At scale $r_n = 2^{-n}$ there are $O(r_n^{-2})$ grid balls in a fixed compact set. Choose q large and γ small so that $(p_\gamma - 1/2)q - q^2\xi^2 - 2 > 0$. Summing (2.5), divided by $r_n^{q/2}$, proves a moment bound for the compact 1/2-Hölder constant. Every pair at distance comparable to r_n lies in one of the enlarged grid balls. Scaling from B_1 to B_{2^m} and summing with a sufficiently large polynomial weight gives (2.2). Only finitely many fixed moment orders have been used.

The occupation time of any Euclidean ball is at most the time elapsed between its first and last visits. Geodesic additivity therefore gives

$$\mu(B_r(z)) \le \frac{\operatorname{diam}_{\widehat{D}_h}(B_r(z))}{\widehat{D}_h(0,1)} \le \min\{1, \mathcal{A} r^{1/2}\}, \qquad \mathcal{A} \le \frac{C C_* (1+\mathcal{R})^K}{\widehat{D}_h(0,1)}. \tag{2.6}$$

The diameter in the first fraction is for unrestricted distances. This argument remains valid if the path visits a ball repeatedly. Layer cake gives

$$\iint \log^+ \frac{1}{|z-w|}\, \mu(\mathrm{d}z)\mu(\mathrm{d}w) \le 2(1 + \log^+ \mathcal{A}).$$

The remaining terms of (1.2) are bounded above by $2\log\mathcal{R}$. Hence

$$\|\mu\|_G^2 \le C(1 + \log\mathcal{R} + \log^+ C_* + \log^+ \widehat{D}_h(0,1)^{-1}). \tag{2.7}$$

One fixed positive moment of each variable on the right, followed by Hölder's inequality, proves (2.3) at one common small-γ threshold. □

2.3 Gaussian differentiation and growing moments

For a smooth bounded Dirichlet function f, Weyl scaling and uniqueness give

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \log \widehat{D}_{h+tf}(0,1) = \xi\mu(f). \tag{2.8}$$

Indeed, testing on the original geodesic gives one derivative inequality; testing on the perturbed geodesic gives the other. The deterministic metric bounds $e^{-\xi|t|\|f\|_\infty}\widehat{D}_h \le \widehat{D}_{h+tf} \le e^{\xi|t|\|f\|_\infty}\widehat{D}_h$ imply compactness of those perturbed geodesics. Each limit is an original geodesic, hence is P by uniqueness. Their normalized length occupations converge as well, by the same uniform length comparison and the geodesic clock.

For completeness, choose a Dirichlet-orthonormal basis (e_j) of smooth bounded functions with zero normalizing-circle average. Along each Gaussian coordinate, the logarithmic distance is Lipschitz with constant $\xi\|e_j\|_\infty$. Cameron–Martin absolute continuity and Fubini give the almost everywhere derivative (2.8). For $F_n = \mathbb{E}[\log\widehat{D}_h(0,1) \mid a_1, \dots, a_n]$,

$$\partial_j F_n = \mathbb{E}[\xi\mu(e_j) \mid a_1, \dots, a_n], \qquad \sum_j |\mu(e_j)|^2 = \|\mu\|_G^2 .$$

Apply the finite-dimensional Gaussian inequalities to these conditional expectations, then pass to the limit by martingale convergence. The energy identity follows first for smoothed measures, then from (2.6).

Gaussian rotation proves the dimension-free estimate

$$\|F - \mathbb{E}F\|_{L^q} \le C\sqrt{q}\, \|\|DF\|\|_{L^q}, \qquad q \ge 2. \tag{2.9}$$

To see this in finite dimensions, integrate $F(X\cos t + Y\sin t)$ for $0 \le t \le \pi/2$, where X, Y are independent standard Gaussian vectors. The rotated velocity is independent of the rotated position. Its conditional scalar Gaussian qth moment is at most $C\sqrt{q}$ times the gradient norm. Minkowski's inequality proves (2.9), and the preceding conditional projections give the general case.

By (2.3) and (2.8),

$$\left\|\log \widehat{D}_h(0,1) - \mathbb{E}\log \widehat{D}_h(0,1)\right\|_{L^q} \le C\xi q, \qquad q \ge 2, \tag{2.10}$$

at a common threshold. Since 0 is a median of $\log \widehat{D}_h(0,1)$, its mean has absolute value at most $C\xi$. Summing the exponential series consequently gives

$$\mathbb{E}e^{c|\log \widehat{D}_h(0,1)|/\xi} \le C, \qquad \sup_{|q|\le c/\xi} \mathbb{E}\widehat{D}_h(0,1)^q \le C. \tag{2.11}$$

This step, rather than an interchange of fixed-moment thresholds, supplies the growing moments used below.

Lemma 2.2. *For each fixed R, put $\Delta_R = \sup_{u,v\in B_R} |\widehat{D}_h(u,v) - |u-v||$. There are constants $C_R, C_0, c > 0$ such that*

$$\|\Delta_R\|_{L^q} \le C_R\xi q, \qquad C_0\log(1/\xi) \le q \le c/\xi. \tag{2.12}$$

In particular, for each fixed K,

$$\mathbb{P}[\Delta_R > C_{R,K}\xi\log(1/\xi)] \le C_{R,K}\xi^K, \tag{2.13}$$

and for fixed sufficiently small $d_0 > 0$, $\mathbb{P}[\Delta_R > d_0] \le e^{-c_{R,d_0}/\xi}$.

Proof. For a, b in a fixed compact set, $r = |b-a|$, affine covariance gives

$$\log \widehat{D}_h(a,b) = p_\gamma \log r + \xi h_r(a) + X,$$

where X has the marginal law of $\log \widehat{D}_h(0,1)$ and $\operatorname{Var} h_r(a) \le C_R + \log^+(1/r)$. Thus

$$\left\|\log \widehat{D}_h(a,b) - p_\gamma \log r\right\|_{L^q} \le C_R\xi\{q + \sqrt{q(1+\log^+(1/r))}\}.$$

Using $|e^x - 1| \le |x|e^{|x|}$, Cauchy–Schwarz, and (2.11) shows, after decreasing c, that

$$\left\|\widehat{D}_h(a,b) - r^{p_\gamma}\right\|_{L^q} \le C_R\xi r^{p_\gamma - C_R q\xi^2}\{q + \sqrt{q(1+\log^+(1/r))}\} \le C_R\xi q \tag{2.14}$$

for $2 \le q \le c/\xi$. The last inequality follows since the power of r is bounded away from zero.

We also need a Hölder constant at these moment orders. Choose small $c_1 > 0$, put $q_1 = c_1/\xi$, and set $s_\gamma = p_\gamma - (c_1 + 3/c_1)\xi$. Affine covariance and (2.11) yield $\mathbb{E}\widehat{D}_h(u,v)^{q_1} \le C_R|u-v|^{p_\gamma q_1 - q_1^2\xi^2}$. At dyadic scale 2^{-n} there are $O_R(2^{2n})$ neighboring grid edges, and

$$(p_\gamma - s_\gamma)q_1 - q_1^2\xi^2 - 2 = 1.$$

The sum over scales is therefore finite. Dyadic parent-chain triangle inequalities extend it from neighbors to all pairs. Since $1/2 < s_\gamma < 1$ for sufficiently small γ (choose c_1 small in view of (1.9)), the chaining constant is bounded. The resulting compact Hölder constant has bounded L^{q_1} norm.

Take a grid of mesh ξ^4 in B_{R+1}. It has $O_R(\xi^{-16})$ ordered pairs. Its maximum error in (2.14) has L^q norm at most $C_R\xi q$ when $q \ge C_0\log(1/\xi)$. Moving endpoints to the grid costs $O_{L^q}(\xi^2)$ by the preceding Hölder bound. Finally $\sup_{0\le r\le 2R+2}|r^{p_\gamma} - r| \le C_R|p_\gamma - 1| \le C_R\xi$. This proves (2.12). Markov's inequality at a suitable multiple of $\log(1/\xi)$ proves (2.13); at a sufficiently small multiple of $1/\xi$ it proves the last assertion. □

3 A residual variance depending on the unknown height

Lemma 3.1. *Suppose a continuous geodesic metric D has uniform additive Euclidean error $\Delta \le d_0$ on B_3, where d_0 is a sufficiently small numerical constant. Parametrize its 0-to-1 geodesic P at constant speed on $[0,1]$, and suppose $\mu(B_u(z)) \le Au^s$ for $u \le 1$, with $s > 0$. Then*

$$\|\mu - \lambda\|_G^2 \le C_s M + C_s \Delta(1 + \log^+ A + |\log \Delta|), \qquad M = \max_t \operatorname{dist}(P(t), [0,1]). \tag{3.1}$$

Proof. A first exit from B_2 would cost more than $D(0,1)$, so P stays in B_2. Geodesic additivity gives $|P(t)| + |P(t) - 1| \le 1 + 3\Delta$, and hence $M \le C\sqrt{\Delta}$. Also $||P(t)| - t| \le 2\Delta$, so

$$\sup_t |P(t) - t| \le 2(M + \Delta) =: r.$$

The first/last-visit argument gives the additional bound $\mu(B_u(z)) \le C(u + \Delta)$. The zero mass of $\mu - \lambda$ cancels the normalization terms of G. Its energy is the integral on $[0,1]^2$ of

$$K(P(t), P(v)) - K(P(t), v) - K(t, P(v)) + K(t, v), \qquad K(z, w) = -\log|z - w|. \tag{3.2}$$

On $|t - v| > 4r$, the mixed Hessian bound gives absolute value at most $Cr^2/|t - v|^2$, whose integral is $O(r)$. On $|t - v| \le 4r$, subtract the common constant $\log(1/r)$ from all four kernels. Their negative parts are bounded. The straight self-interaction is $O(r)$; the positive mixed terms occur with a minus sign and can be discarded in an upper bound. By layer cake, the remaining self-interaction is bounded by

$$\int_0^\infty \min\{Cr, C(re^{-a} + \Delta), Ar^s e^{-sa}\}\, \mathrm{d}a \le Cr + C_s\Delta(1 + \log^+ A + |\log \Delta|).$$

Use $\min(x + y, z) \le x + \min(y, z)$ for the last inequality. Truncation of the logarithmic singularities, followed by the Frostman bound, justifies each integral and its limit. Since $r \le C(M + \Delta)$, this proves (3.1). □

Let $m = \mathbb{E}M_\gamma$, $L = \log(1/\xi)$, and define the centered residual

$$\mathcal{Z}(h) = \log \widehat{D}_h(0,1) - \mathbb{E}\log \widehat{D}_h(0,1) - \xi(h, \lambda). \tag{3.3}$$

The Gaussian pairing with λ is defined since λ has finite logarithmic energy.

Lemma 3.2. *Uniformly for sufficiently small γ,*

$$\mathbb{E}\mathcal{Z}(h)^2 \le C\xi^2(m + \xi L^2). \tag{3.4}$$

Proof. Gaussian differentiation as above gives $D\mathcal{Z} = \xi(\mu - \lambda)$ in the Dirichlet dual space. On the fixed localization event $\{\Delta_3 \le d_0\}$, use Lemma 3.1 with $s = 1/2$ and the compact Hölder constant divided by $\widehat{D}_h(0,1)$. Its $\log^+$ has fixed finite moments. By Lemma 2.2, $\|\Delta_3\|_{L^2} \le C\xi L$; also $x|\log x| \le Lx + \xi L$ for $0 \le x \le d_0 < 1$. Cauchy–Schwarz gives

$$\mathbb{E}[\|\mu - \lambda\|_G^2 \mathbf{1}_{\{\Delta_3 \le d_0\}}] \le C(m + \xi L^2).$$

The complement has probability $e^{-c/\xi}$. Its contribution is exponentially small by (2.3) and the bound

$$\|\mu - \lambda\|_G^2 \le 2\|\mu\|_G^2 + 2\|\lambda\|_G^2.$$

Gaussian Poincaré proves (3.4). In particular, this estimate has not assumed the desired bound on m. □

For the conformal argument we henceforth subtract the field average on $\partial B_{10}(0)$ instead of $\partial\mathbb{D}$. Distances acquire a common random factor, while P, μ, and $\mathcal{Z}$ are unchanged. The extra average is a centered Gaussian of fixed variance. Equations (2.12)–(3.4), with changed constants, therefore remain valid. This normalization places the reference circle strictly outside all the conformal perturbations below.

4 Analytic probes and a local Gaussian coupling

4.1 Clock geometry and reconstruction

Write $P(t) = x(t) + \mathrm{i}y(t)$. On $\{\Delta_3 \le d_0\}$, subtracting the squares of $|P(t)| = t + O(\Delta_3)$ and $|P(t) - 1| = 1 - t + O(\Delta_3)$ gives

$$|x(t) - t| \le C\Delta_3, \qquad |y(t) - y(u)| \le C\sqrt{|t-u|\Delta_3} + C\Delta_3, \qquad M \le \|y\|_\infty + C\Delta_3. \tag{4.1}$$

For the second inequality use $|P(t) - P(u)| \le |t-u|(1+\Delta_3) + \Delta_3$ and the first one. No graph representation of P is involved.

Choose an even smooth function χ, equal to 1 on $[-1,1]$ and supported in $[-2,2]$, and put $\rho(s) = \chi(s) - \chi(2s)$, $\ell_j = 2^{-j}$. For $a \in [0,1]$ define the entire function

$$H_{j,a}(z) = -\frac{2}{\ell_j} \sum_{k \ge 1} \frac{\rho(k\ell_j)\sin(k\pi a)\sin(k\pi z)}{(k\pi)^2}. \tag{4.2}$$

The sum is finite, $H_{j,a}$ is real on the real line, and $H_{j,a}(0) = H_{j,a}(1) = 0$. Define

$$B_{j,a} = \ell_j \int_0^1 y(t) H''_{j,a}(t)\,\mathrm{d}t. \tag{4.3}$$

The function $a \mapsto B_{j,a}$ is the sine-frequency band of y with multiplier $\rho(k\ell_j)$.

Lemma 4.1. *For each j, one can choose a deterministic set $\mathcal{A}_j$ of at most C/ℓ_j sample points such that, for every continuous $y : [0,1] \to \mathbb{R}$ with $y(0) = y(1) = 0$,*

$$\|y\|_\infty \le 2\sum_{j=0}^{J} \max_{a \in \mathcal{A}_j} |B_{j,a}| + C\omega_y(\ell_J). \tag{4.4}$$

The probes satisfy $\|H\|_{L^\infty(\mathbb{R})} \le C$, $\int_0^1 |H| \le C\ell$, and $\int_0^1 |H'|^2 \le C/\ell$. On every fixed multiple of the complex strip of width ℓ, each $H^{(k)}$ is bounded by $C_k\ell^{-k}$ times a bounded localized envelope whose integral over $[0,1]$ is at most $C_k\ell$.

Proof. The sine-product formula writes H as the difference of two period-two Fourier kernels with a smooth annular multiplier. Poisson summation expresses each as a periodization of a Schwartz function on scale ℓ. Complex displacement of size $O(\ell)$ multiplies the rescaled multiplier by $e^{-\pi s\,\mathrm{Im}(z)/\ell}$, which preserves its bounded smooth seminorms. This proves the asserted derivative envelopes, including their integrals. Orthogonality also gives directly

$$\int_0^1 |H'|^2 = \frac{2}{\ell^2} \sum_{k \ge 1} \frac{\rho(k\ell)^2 \sin^2(k\pi a)}{(k\pi)^2} \le C/\ell.$$

The band has trigonometric degree at most $2/\ell_j$. Bernstein's inequality permits its supremum to be bounded by twice the maximum on a grid of mesh $\ell_j/32$. Finally $\sum_{j=0}^{J} \rho(k\ell_j) = \chi(k\ell_J)$ for integers $k \ge 1$. The low-pass kernel is an approximate identity with bounded L^1 norm and first absolute moment $O(\ell_J)$. Apply it to the continuous odd period-two extension of y to obtain the remainder in (4.4). □

4.2 The coupling lemma

The next estimate concerns Gaussian fields, independently of the metric.

Lemma 4.2. *Let H be one of the probes at scale $\ell \le 1$, and let $\phi(z) = z + \mathrm{i}\theta H(z)$, where $|\theta|/\ell$ is sufficiently small. There are nested rectangles $V \Subset U$ around $[0,1]$, with width and vertical separation comparable to ℓ and fixed horizontal margins, and a coupling with a canonically normalized whole-plane GFF $\bar{h}$ such that on U*

$$h \circ \phi^{-1} = \bar{h} + K, \qquad \Big\| \sup_V |K| \Big\|_{L^2} \le C \frac{|\theta|}{\ell} (1 + \log(1/\ell)). \tag{4.5}$$

Here K is harmonic. For each prescribed probe the original h can be kept fixed; the reference $\bar{h}$ is allowed to depend on the probe.

Proof. Extend ϕ smoothly to a map

$$\Phi(z) = z + \mathrm{i}\theta \chi_x(\operatorname{Re} z)\chi_y(\operatorname{Im} z/\ell) H(z),$$

using cutoffs supported within the analytic strip and strictly inside B_{10}. The extension equals ϕ on a narrower strip and satisfies $\|D\Phi - I\|_\infty \le C|\theta|/\ell =: C\varepsilon$. It is a global diffeomorphism for small ε. Its inverse agrees with ϕ^{-1} on U.

The Cameron–Martin energy of $h \circ \Phi^{-1}$ is $\mathcal{E}_\theta(f) = \mathcal{E}(f \circ \Phi)$, where $\mathcal{E}(f) = \|f\|_\nabla^2$. Change of variables gives

$$(1 - C\varepsilon)\mathcal{E}(f) \le \mathcal{E}_\theta(f) \le (1 + C\varepsilon)\mathcal{E}(f).$$

The coefficient matrix of this form is exactly the identity on U, by conformality. Consequently the decomposition into the zero-boundary Dirichlet space on U and its harmonic orthogonal complement is orthogonal for both energies; the energies agree on the first summand. Use one common zero-boundary GFF on U. On the harmonic complement write $\mathcal{E}_\theta(f, g) = \mathcal{E}(Bf, g)$. Then $\|B^{-1/2} - I\|_{\mathrm{op}} \le C\varepsilon$. Applying I and $B^{-1/2}$ to one harmonic Gaussian coordinate sequence couples the harmonic parts H_0, H_θ so that, for every bounded evaluation functional T,

$$\operatorname{Var} T(H_\theta - H_0) \le C\varepsilon^2 \operatorname{Var} T(H_0). \tag{4.6}$$

These are Gaussian series against bounded evaluation functionals, not Hilbert-space-valued standard Gaussian vectors. Conversely one can whiten the harmonic part of the given $h \circ \Phi^{-1}$; this keeps the original h fixed for each of the finitely many probes used later.

On V, the harmonic variance is at most $C(1 + \log(1/\ell))$: subtract the zero-boundary Green function from the whole-plane kernel, and use that the conformal radius is at least $c\ell$. Interior harmonic estimates similarly bound its canonical increment metric by $C\sqrt{1 + \log(1/\ell)} \min\{1, |z - w|/\ell\}$. By (4.6), the corresponding bounds for $K = H_\theta - H_0$ are multiplied by $C\varepsilon$. A Euclidean net gives covering number at most

$$C\ell^{-2} \left(\frac{C\varepsilon\sqrt{1 + \log(1/\ell)}}{u} \right)^2$$

at canonical-metric radius u below the diameter. Integrating the square root of its logarithm gives expected supremum $C\varepsilon(1 + \log(1/\ell))$; the same bound in L^2 follows from the Gaussian supremum concentration inequality. Equivalently this follows by telescoping dyadic Gaussian nets. This proves (4.5).

The common zero-boundary field is independent of the pair of harmonic parts. Conditional on that pair, ordinary Weyl scaling applies to their deterministic smooth difference. Thus the coupling also compares internal metric lengths on compact subsets of V. It does not apply a deterministic-shift theorem to an unconditioned field-dependent shift. □

5 Critical upper bound

Set $\delta = \xi^{2/3}$ and $L = \log(1/\xi)$. Fix a large numerical K_0, for example $K_0 = 20$. By Lemmas 2.1 and 2.2, there is an event $\mathcal{G}$ such that

$$\mathbb{P}[\mathcal{G}^c] \le C\xi^{K_0}, \qquad \Delta_3 \le C\xi L, \qquad M \le C\xi^{1/2}L^{1/2} \quad \text{on } \mathcal{G}. \tag{5.1}$$

Here and below $M = M_\gamma$. The second height bound is just the ellipse bound; it contains no critical-scale input. Uniform escape moments imply $\mathbb{E}[M\mathbf{1}_{\mathcal{G}^c}] = o(\delta)$.

Choose J so that $\ell_J \asymp \xi^{3/8}$. By (4.1) and (4.4), on $\mathcal{G}$,

$$M \le 2\sum_{j=0}^{J} \max_{a\in\mathcal{A}_j} |B_{j,a}| + C\xi^{11/16}L^{1/2} + C\xi L, \tag{5.2}$$

whose remainder is $o(\delta)$.

5.1 A single-probe estimate

Fix a probe $H = H_{j,a}$, with scale ℓ, and use $\phi(z) = z + \mathrm{i}\theta H(z)$ and Lemma 4.2. For the parameters chosen below, the path and its image lie in the inner analytic strips on $\mathcal{G}$. Mapping the original geodesic by ϕ, then using it as a competitor for the *global* reference distance, gives

$$\log \widehat{D}_{\bar h}(0,1) - \log \widehat{D}_h(0,1) \le \log \int_0^1 \exp\{p_\gamma \log|\phi'(P(t))| - \xi K(\phi(P(t)))\}\,\mathrm{d}t. \tag{5.3}$$

The sign follows from $h\circ\phi^{-1} + Q\log|(\phi^{-1})'| = \bar h + K - Q\log|\phi'|\circ\phi^{-1}$. Only the original path is localized. No minimizing property of its image for the reference metric is asserted.

Let $G_\theta = (h, (\phi^{-1})_*\lambda - \lambda)$. Pairing (4.5) with the real segment and subtracting the linear Gaussian terms in (3.3) yields

$$\xi G_\theta + \mathcal{Z}(\bar h) - \mathcal{Z}(h) \le \log \int_0^1 e^{p_\gamma \log|\phi'(P(t))|}\,\mathrm{d}t + 2\xi \sup_V |K|. \tag{5.4}$$

The two residuals have the same marginal law. They need not be independent.

The measure $(\phi^{-1})_*\lambda$ is coupled to λ by displacement at most $C|\theta|$, and both measures have a uniform linear ball-mass bound. The four-kernel proof of Lemma 3.1, now with no additive Δ term, gives

$$\mathbb{E}G_\theta^2 \le C|\theta|. \tag{5.5}$$

More explicitly, split the parameter square at separation $C|\theta|$. The far part is bounded by the mixed Hessian as before; on the near part, the layer-cake bound is $\min\{C|\theta|, C|\theta|e^{-a}\}$, whose integral is $O(|\theta|)$. Uniform bi-Lipschitzness of ϕ^{-1} along the segment supplies the linear ball-mass bound.

The reality of H on the real axis and Taylor expansion give

$$\log|1 + \mathrm{i}\theta H'(x+\mathrm{i}y)| = -\theta y H''(x) + \frac{\theta^2}{2}H'(x)^2 + \text{error}.$$

Using the localized derivative envelopes in Lemma 4.1 and $|x(t) - t| \le C\Delta_3$ gives

$$\log \int_0^1 e^{p_\gamma \log|\phi'(P(t))|}\,\mathrm{d}t \le -p_\gamma \frac{\theta}{\ell} B_{j,a} + C\frac{\theta^2}{\ell} + \mathrm{Err}, \tag{5.6}$$

where

$$|\mathrm{Err}| \le C\left(\frac{|\theta|M\Delta_3}{\ell^2} + \frac{|\theta|M^3}{\ell^3} + \frac{\theta^2M^2}{\ell^3} + \frac{|\theta|^3M}{\ell^3} + \frac{\theta^4}{\ell^3}\right). \tag{5.7}$$

Here is the error bookkeeping. Expanding in y first leaves pointwise errors bounded by the localized envelope times $C(|\theta||y|^3 + \theta^2y^2 + |\theta|^3|y| + \theta^4)/\ell^4$. Its integral gains a factor ℓ. Replacing $H''(x(t))$ by $H''(t)$ costs $CM\Delta_3/\ell^2$. The quadratic energy of H' is $O(1/\ell)$. Finally, $\log\int e^U \le \int U + C\int U^2$ when $\|U\|_\infty$ is bounded; its error is at most $C(\theta^2M^2+\theta^4)/\ell^3$, already included above.

Fix $\beta = 1/8$. For a threshold R with

$$\delta \le R \le \xi^{1/2}L, \tag{5.8}$$

put $r_j = c_\beta R\ell_j^\beta$, where c_β is chosen so that $2\sum_{j\ge0} c_\beta\ell_j^\beta < 1/2$. For the two signs of $B_{j,a}$ choose respectively $\theta = \pm c_* r_j$, with small fixed $c_* > 0$. Uniformly over these choices,

$$\frac{M}{\ell} \le C\xi^{1/8}L^{1/2}, \qquad \frac{|\theta|}{\ell} \le C\xi^{11/64}L,$$

so all analytic-domain requirements hold for small ξ. Dividing the five terms in (5.7) by θ^2/ℓ gives bounds by powers of L times, respectively,

$$\xi^{79/192}, \quad \xi^{7/192}, \quad \xi^{1/4}, \quad \xi^{19/64}, \quad \xi^{11/32}. \tag{5.9}$$

For the first two use the lower endpoint in (5.8), and for the last two use its upper endpoint. All tend to zero. They also imply $\|p_\gamma \log|\phi'(P)|\|_\infty = o(1)$.

Choose c_* small enough that the deterministic quadratic term in (5.6) is absorbed by half the linear term on $\{\pm B_{j,a} > r_j\}$. The remaining error is absorbed for small ξ. Equations (5.4)–(5.5), Lemma 3.2, and Chebyshev's inequality yield

$$\mathbb{P}[|B_{j,a}| > r_j, \mathcal{G}] \le C\left(\frac{\xi^2\ell_j^2}{r_j^3} + \frac{\xi^2(m+\xi L^2)\ell_j^2}{r_j^4} + \frac{\xi^2L^2}{r_j^2}\right). \tag{5.10}$$

The three terms arise from G_θ, the two residuals, and the harmonic coupling error. No independence between any of these variables and the path is required.

5.2 Summing the probes and closing the bound

By (5.2), the event $\{M > R, \mathcal{G}\}$ implies a probe event in (5.10), for sufficiently small ξ uniformly in (5.8). There are at most $C\ell_j^{-1}$ probes at level j. The first two level sums have powers $\ell_j^{1-3\beta}$ and $\ell_j^{1-4\beta}$ and are summable. The last is at most $C\ell_J^{-1-2\beta} \le C\xi^{-15/32}$. Therefore

$$\mathbb{P}[M > R, \mathcal{G}] \le C\left(\frac{\xi^2}{R^3} + \frac{\xi^2(m+\xi L^2)}{R^4} + \frac{\xi^{49/32}L^2}{R^2}\right). \tag{5.11}$$

For a fixed sufficiently large A, integrate this inequality from $A\delta$ to $\xi^{1/2}L$. On $\mathcal{G}$ the height is below the upper endpoint by (5.1); the contribution on $\mathcal{G}^c$ is $o(\delta)$. Hence

$$\begin{aligned} m &\le A\delta + C\delta A^{-2} + CmA^{-3} + O(\xi L^2) + O(\xi^{83/96}L^2) + o(\delta) \\ &\le A\delta + C\delta A^{-2} + CmA^{-3} + o(\delta). \end{aligned} \tag{5.12}$$

The quantity m is already finite by Lemma 2.1. Choosing A so that $CA^{-3} < 1/2$ and absorbing proves

$$\mathbb{E}M_\gamma \le C\xi^{2/3}. \tag{5.13}$$

Substitute this bound in (5.11), take $R = A\xi^{2/3}$ for fixed $A \ge 1$, and let $\xi \to 0$. The harmonic error tends to zero since $49/32 - 4/3 = 19/96 > 0$; the additional deterministic residual error also vanishes. Thus

$$\limsup_{\gamma\to 0} \mathbb{P}[M_\gamma > A\xi^{2/3}] \le C(A^{-3} + A^{-4}). \tag{5.14}$$

By (1.9), these prove (1.3) and (1.4).

6 A quantitative lower tail from Möbius bending

The following proposition is a direct prelimit estimate. In particular, it requires none of the upper-bound argument.

Proposition 6.1. *There are numerical $C < \infty$ and $r_* > 0$ such that for $\gamma \in (0,2)$ and $0 < r \le r_*$,*

$$\mathbb{P}[M_\gamma \le r] \le C(Qr^{3/2})^{2/5}. \tag{6.1}$$

The bound is interpreted as trivial when its right side exceeds one.

Proof. Let $T_r = \{z : \operatorname{dist}(z,[0,1]) \le r\}$ and set

$$\phi_\theta(z) = \frac{e^{\mathrm{i}\theta}z}{1 + (e^{\mathrm{i}\theta} - 1)z}, \qquad \psi_\theta = \phi_{-\theta}.$$

These maps fix $0, 1$. They preserve the vertical line $\operatorname{Re} z = 1/2$, and, with $t = \tan(\theta/2)$,

$$\phi_\theta(1/2 + \mathrm{i}y) = \frac{1}{2} + \mathrm{i}\frac{y + t/2}{1 - 2ty}. \tag{6.2}$$

For an integer $N \ge 1$, take $\theta_k = 64kr$, $0 \le k \le N$, and write $R = Nr$. Assume R is below a small numerical constant. The intersections of $\phi_{\theta_k}(T_r)$ with the midline are contained in intervals of radius $6r/5$ centered at $\frac{1}{2}\tan(\theta_k/2)$. These centers are separated by at least $16r$, and the derivative in y in (6.2) is at most $6/5$ on $|y| \le r$. Thus the intersections are pairwise disjoint. The inverse map preserves the same line, so this computes the entire intersection, not just part of it.

All these tubes lie in one open rectangle

$$U_R = \{-1/4 < \operatorname{Re} z < 5/4, \quad |\operatorname{Im} z| < KR\},$$

for a fixed K. Every map and its inverse is conformal on a fixed disk containing this rectangle and the unit circle when R is small.

We first record the exact local GFF law. For a Möbius map $\psi(z) = (az + b)/(cz + d)$,

$$\log|\psi(z) - \psi(w)| = \log|ad - bc| + \log|z - w| - \log|cz + d| - \log|cw + d|.$$

Consequently $G(\psi(z), \psi(w)) - G(z, w)$ is a sum of a function of z, a function of w, and a constant. Subtracting circle averages in both variables cancels this difference. Thus, locally on the fixed disk,

$$\widetilde{h}_k = h \circ \psi_{\theta_k} - (h \circ \psi_{\theta_k})_1(0)$$

has the canonical normalized GFF law. Circle averages on the image circle are legitimate finite-energy Gaussian pairings. This argument uses no assertion across a pole of the Möbius map.

Choose smooth cutoffs $\eta(x)$, equal to one on $[-1/4, 5/4]$ and supported in $[-1/2, 3/2]$, and $\chi_0(y)$, equal to one on $[-1, 1]$ and supported in $[-2, 2]$. Set

$$g_k(z) = Q\eta(\operatorname{Re} z)\chi_0(\operatorname{Im} z/(KR)) \log|\psi'_{\theta_k}(z)|. \tag{6.3}$$

Its support avoids all poles, and it equals $Q \log|\psi'_{\theta_k}|$ on U_R. Since

$$\log|\psi'_\theta(z)| = -2 \operatorname{Re} \log(1 + (e^{-\mathrm{i}\theta} - 1)z),$$

on the cutoff support its absolute value is at most $C(|\theta|R + \theta^2) \le CR^2$, and its gradient is at most $C|\theta| \le CR$. The vertical cutoff derivative costs R^{-1}, so all first derivatives of (6.3) are at most CQR. The support has area $O(R)$; therefore

$$\|g_k\|_\nabla^2 \le CQ^2R^3. \tag{6.4}$$

Cameron–Martin relative entropy and Pinsker's inequality imply total variation distance at most $CQR^{3/2}$ between the normalized GFF and its shift by $g_k - (g_k)_1(0)$. Additive constants have no effect on the geodesic events below.

Let B_k be the event that the internal metric in the *same* domain U_R has a unique 0-to-1 geodesic and that this geodesic lies in $\phi_{\theta_k}(T_r)$. The events $B_0, \dots, B_N$ are disjoint: the common unique curve must cross the midline, whereas the corresponding intervals there are disjoint. Existence and uniqueness of an internal geodesic are part of the event, not an extra theorem being assumed.

On $A = \{M_\gamma \le r\}$, the original global geodesic lies in $T_r \subset \psi_{\theta_k}(U_R)$. By (2.1), its image is an internal geodesic in U_R for $h \circ \psi_{\theta_k} + Q \log|\psi'_{\theta_k}|$. It is unique: another internal minimizer would pull back to a distinct global original geodesic of the same length. By locality and Weyl scaling,

$$A(h) \quad \Longrightarrow \quad B_k(\widetilde{h}_k + g_k).$$

We have not claimed that the image stays globally minimizing after cutting off the conformal drift. The internal event is precisely what avoids that problem.

Put $p = \mathbb{P}(A)$. Local equality of the Gaussian laws and (6.4) give $\mathbb{P}[B_k(h)] \ge p - CQ(Nr)^{3/2}$. Summing the disjoint events yields

$$p \le \frac{1}{N+1} + CQ(Nr)^{3/2}, \qquad Nr \le R_0. \tag{6.5}$$

When $b = Qr^{3/2}$ is small, choose $N \asymp b^{-2/5}$. Both terms in (6.5) are at most $Cb^{2/5}$. The geometric restriction holds for sufficiently small universal r_*, since $Nr \le CQ^{-2/5}r^{2/5}$ and $Q \ge 2$. For the remaining b, increase C and use the trivial bound one. This proves (6.1). □

7 Consequences and completion of the proof

Apply Proposition 6.1 with $r = a\gamma^{2/3}$. Since

$$Qr^{3/2} = (2 + \gamma^2/2)a^{3/2},$$

reducing γ_0 so that $\gamma_0^{2/3} \le r_*$ proves (1.5). For $X_\gamma = M_\gamma/\gamma^{2/3}$ and $r \in (0, 3/5)$, layer cake gives

$$\mathbb{E}X_\gamma^{-r} \le 1 + r\int_1^\infty t^{r-1}\mathbb{P}[X_\gamma < t^{-1}]\,\mathrm{d}t \le 1 + Cr\int_1^\infty t^{r-1-3/5}\,\mathrm{d}t < \infty,$$

uniformly in γ. This proves (1.7). Combining the lower bound with (1.3) or (1.4) gives (1.6).

For clarity about subsequential limits, if $X_{\gamma_n} \Rightarrow X$, then for every $a > 0$ the open-set form of Portmanteau gives

$$\mathbb{P}[X < a] \le \liminf_n \mathbb{P}[X_{\gamma_n} < a] \le C a^{3/5} \quad (a \le 1).$$

Letting $a \downarrow 0$ shows $\mathbb{P}[X = 0] = 0$; upper tightness gives a finite limit. Finally, Markov's inequality and (1.5) give

$$\mathbb{P}[M_\gamma > \gamma^{2/3-\varepsilon}] \le C\gamma^{\varepsilon}, \qquad \mathbb{P}[M_\gamma < \gamma^{2/3+\varepsilon}] \le C\gamma^{3\varepsilon/5},$$

which proves (1.8) and completes Theorem 1.1.

Remark 7.1. The two-thirds power is established here by quantitative bounds for the actual geodesic, not only by a scaling balance. The lower proof identifies the conformal bending cost Q^2R^3, which becomes order one when $R \asymp \gamma^{2/3}$. The upper proof matches that scale without a logarithmic loss by summing localized analytic probes and absorbing the unknown mean height in (5.12). Identifying the law of a subsequential limit is a separate question and is not a prerequisite for the theorem.

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